

DSA

Lab # 8

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Question No. 1 :-

Convert infix $\rightarrow$ Postfix

1) Infix

$$A + B * (C^D - E)^{(F + G * H)} - I$$

Postfix

$$ABCD^E - FG H * + ^ * + I -$$

2) Infix

$$A + (B * C - (D / E^F) * G) * H$$

Post fix

$ABC * DEF ^{1/4} * - H * +$

Question No. 2:

Convert Infix $\rightarrow$ Prefix

1) Infix

$$A + (B * (C + (D * F)))$$

Prefix

$$+ A * B + C * DE$$

2) Infix

$$A * (B + C * (D - E)) + F / G$$

Prefix

$$+ * A + B * C - DE / FG$$

Question No. 3

Evaluate the following expressions:

1) $56 + 2 * 34 + -$

$$5 + 6 = 11$$

$$11 * 2 = 22$$

$$3 + 4 = 7$$

$$22 - 7 = -15$$

$$2) - * 45 + 23$$

$$4 \times 5 = 20$$

$$2 + 3 = 5$$

$$20 - 5 = 15$$

$$3) + * 234$$

$$2 \times 3 = 6$$

$$6 + 4 = 10$$

$$4) 93 * 42/-$$

$$9 \times 3 = 27$$

$$4/2 = 2$$

$$27 - 2 = 25$$